🌿 GreenSight MVP — In-Depth Overview

🎯 Core Goal

To build an **AI + LiDAR-powered vegetation monitoring system** that:

- Reads and processes **3D LiDAR data** (forest structure)
- Analyzes **NDVI (Normalized Difference Vegetation Index)** from satellite imagery
- Uses an **AI model (CNN)** to classify vegetation health
- Generates visual and numerical insights for **forestry or agriculture**

🧠 In-Depth Functional Flow

1. LiDAR Data Processing (3D Forest Mapping)

- **File:** `src/preprocess_lidar.py`
- **What Happens:**
  - Loads `.las` LiDAR point cloud file
  - Extracts **X, Y, Z coordinates** of millions of vegetation points
  - Computes:
    - 🌲 **Canopy height** (max Z - min Z)
    - 🌳 **Vegetation density** (number of LiDAR points per m²)
  - Displays 3D forest point cloud using **Open3D** viewer

💡 *Why:* This simulates how drones or LiDAR sensors "see" vegetation structure.

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2. NDVI Extraction (Vegetation Health Indicator)

- **File:** `src/extract_ndvi.py`

- **What Happens:**
    - Loads satellite image (`.tif`)
    - Extracts **Red** and **NIR (Near Infrared)** bands
    - Calculates **NDVI = (NIR - Red) / (NIR + Red)**
    - Generates a vegetation health heatmap (green = healthy, yellow = stressed)

💡 *Why:* NDVI shows how healthy vegetation is — core metric in remote sensing.

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3. AI Model Training (CNN for Vegetation Health)

- **File:** `src/vegetation_model.py`
- **What Happens:**
    - Creates a small **Convolutional Neural Network (CNN)**
    - Simulates training on NDVI image patches
    - Outputs a model that can predict *Healthy* or *Stressed vegetation*

💡 *Why:* The AI model learns vegetation patterns for future real dataset integration.

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4. Visualization & Reporting

- **File:** `src/visualize_results.py`
- **What Happens:**
    - Overlays NDVI map with predicted vegetation health zones
    - Displays classification results visually (like a vegetation dashboard)

💡 *Why:* Converts raw LiDAR/NDVI data into meaningful insights.

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5. Integration and API

- **File:** `backend/app.py`
- **What Happens:**
  - Exposes a **FastAPI** endpoint `/analyze`
  - When called, it runs NDVI analysis and returns:
    - Average NDVI value
    - Overall vegetation health status

💡 *Why:* Makes your system ready for integration with a React dashboard or IoT devices.

⚙️ Step-by-Step to Run the Project

Step 1: Setup Environment

Open your terminal and run:

git clone https://github.com/<your-username>/greensight-mvp.git
cd greensight-mvp
pip install -r requirements.txt

Step 2: Download Test Data

mkdir data
cd data
wget https://opentopography.s3.sdsc.edu/lidar/demonstration/Sonoma_CA.las -O forest_sample.las
wget https://github.com/chrieke/awesome-sentinel/raw/main/sample_data/sentinel_image.tif -O satellite_image.tif
cd ..

Step 3: Run the MVP Pipeline

python src/main.py

✅ **Expected Output Example:**

🌿 GreenSight MVP Pipeline Start
[LiDAR] Canopy Height: 32.45 | Density: 0.0187

[NDVI] NDVI Range: -0.24 - 0.92
[AI] Simulated vegetation health: Healthy
✅ GreenSight MVP Pipeline Completed

Step 4: Run API Backend (Optional)

To expose your model as a web service:

```
uvicorn backend.app:app --reload
```

Then open your browser:

- `http://127.0.0.1:8000` → API Home
- `http://127.0.0.1:8000/analyze` → NDVI health JSON output

✅ Example:

```
{
  "average_ndvi": 0.62
}
```

Step 5: (Optional) Run on Google Colab

1. Upload your repo or open via link
2. Open `/notebooks/greensight_mvp_colab.ipynb`
3. Click **Runtime** → **Run all**
4. Colab will auto-install dependencies, download data, and show NDVI + LiDAR visualization inline.

🔬 End Result

Module	Input	Output	Tech Used
LiDAR Processor	`.las` file	Canopy height + density	laspy, open3d

NDVI Extractor	`.tif` image	Vegetation health map	rasterio, numpy
AI Model	NDVI patch	"Healthy / Stressed"	TensorFlow
Visualization	NDVI data	Colored vegetation map	matplotlib
API Service	HTTP request	JSON vegetation summary	FastAPI

🌱 Real-World Application

Domain	Use Case
🌳 Forestry	Forest health tracking, biomass estimation
🌾 Agriculture	Crop monitoring, yield forecasting
🌍 Environmental Science	Deforestation analysis, ecosystem mapping
🚁 UAV Systems	Edge AI deployment on drone data